

circuit board (PCB) space while enabling higher power densities.

Microchip Technology
www.microchip.com

Memory chip

In a bid to offer a next-generation solution for ultra-responsive low latency operation, Teledyne e2v has announced DDR4T04G72M, a radiation-tolerant DDR4 memory chip



that features 4GB capacity. It is currently validated at 2133MT/s with a

target to increase to 2400MT/s in the near future. It features a 72-bit bus, where 64 bits are dedicated to data and 8 bits to error correction code (ECC). Further, high-reliability manufacturing and radiation-tolerant robustness make it highly suitable for dealing with the rigours of space environments.

The DDR4T04G72M enables increased performance while taking up minimal board space—something that is certain to be of value in highly space-constrained and densely-packed satellite designs. It can be used in conjunction with processors and FPGAs having a DDR4 controller, and is in particular, embedded on the Teledyne e2v space version of the Qormino common compute platform together with the space version of NXP LS1046 quad-core processor (QLS1046-4GB).

Teledyne e2v
www.teledyne-e2v.com

MEMS sensor

Murata has developed a new micro-electro-mechanical system (MEMS) with six degrees of freedom (6DoF) inertial sensor SCHA600 that provides high performance for safety-critical automotive applications.



The sensor enables new advanced driver-assistance systems (ADAS)

functionality and automated driving through data fusion with Global Navigation Satellite System (GNSS) and various sensors such as camera, radar, and lidar.

This new sensor is a single package 6DoF component integrated with GNSS and perception sensors for L2+ and

higher levels of autonomy. It enables centimetre-level-accuracy with respect to vehicle dynamics and position, thus enabling safe and robust autonomous driving. The device guarantees better operation during high mechanical shock and vibration.

It comes in an SOIC housing (32 pins) with dimensions of 18.7mm × 8.5mm × 4.5mm. The product is qualified according to the AEC-Q100 standard and is ISO26262 ASIL-D (Automotive Safety Integrity Level) compliant.

Murata
www.murata.com

MCU

NXP Semiconductors has introduced a new family of ultra-low-power, multi-protocol wireless MCUs, K32W061/41. The new low-power devices complement the recently introduced pin-compatible JN5189/88

(Thread/Zigbee) and QN9090/30 (Bluetooth LE) MCUs and provide original equipment manufacturers (OEMs) with a simple support system for current and emerging smart home and building use cases.



Low-power consumption of today's smart home and Internet of Things (IoT) devices is crucial for achieving high performance from a single-coin cell battery. NXP's K32W061/41 MCU achieves this via multiple low-power modes and low transmit/receive radio power capabilities.

The K32W061/41 wireless micro-controllers are based on Arm Cortex M4 microcontroller core running at 48MHz and include 640kB onboard flash and 152kB SRAM, providing storage space and flexibility for complex applications and software over-the-air (OTA) updates. The optional NFC NTAG provides standardised out-of-band communications for simplifying the pairing process.

NXP Semiconductor
www.nxp.com

Motor control chip

Dialog Semiconductor, a provider of battery management, AC-DC power conversion, Wi-Fi, BLE, and Industrial ICs, has

launched SLG47105, a motor drive CMIC that delivers both configurable logic and configurable analogue with high-voltage outputs.

The device is a part of the company's GreenPAK product family and provides a cost-effective, one-time NVM programmable option for designers to integrate both digital and analogue system functions while minimising component count, board space and power consumption for consumer and industrial motor applications.



The new device is capable of driving two brushed DC motors, a single stepper motor, solenoid, or any other load requiring up to 1.5A RMS current per output. It has an operating voltage of up to 13.2V. Besides protection features such as over-temperature, under-voltage and overcurrent protection, the SLG47105 also includes configurable digital and analogue resources, allowing the user to create a customised protection and motor control scheme with current or voltage regulation, stall detection or soft motor start to enable higher system reliability and more efficient battery usage.

Dialog Semiconductor
www.dialog-semiconductor.com

POWER SUPPLIES

AC power supply

RECOM has announced the extension of its open-frame AC-DC power supply range with the addition of the RACM60-K series. The low-profile 5cm × 7cm design features 90 per cent efficiency across its load range and enables full-power operation from -40°C to +55°C without forced cooling, and up to +85°C with derated or forced air. Input range is from 80V AC to 264V AC and outputs available are 5V, 12V, 15V, 24V, 36V and 48V.



The products are suitable for a wide range of applications with EN 62368-1, IEC 61558-1/-2-16, and IEC/EN 60335-1 certifications for ITE/audio-visual, industrial, and household use, respectively.

The RACM60-K series features full protection against short-circuits, output